

CS5187 Final Exam Review / 期末复习资料

Course / 课程: CS5187: Vision & Image Instructor / 授课教师: Prof. Zhu Zhu

1. Knowledge Framework / 知识框架

Part I: Low-Level Vision / 低层视觉 (lec00-lec05)

Lecture	Topic / 主题
lec00	Introduction to Computer Vision / 计算机视觉导论
lec01	Images & Image Filtering / 图像与图像滤波
lec02	Edge Detection / 边缘检测
lec03	Image Resampling & Interpolation / 图像重采样与插值
lec04	Harris Corner Detection / Harris角点检测
lec05	Feature Invariance / 特征不变性

Part II: Geometry & Appearance / 几何与外观 (lec06-lec10, lec13-lec14)

Lecture	Topic / 主题
lec06	Feature Descriptors & Matching / 特征描述子与匹配
lec07	Image Transformations & Warping / 图像变换与变形
lec08	Image Alignment / 图像对齐
lec09	RANSAC / 随机抽样一致
lec10	Cameras & Projection / 相机与投影
lec13	Binocular Stereo / 双目立体视觉
lec14	Multi-View Stereo / 多视图立体

Part III: Recognition & Deep Learning / 识别与深度学习 (lec15-lec16)

Lecture	Topic / 主题
lec15	Neural Networks / 神经网络
lec16	CNNs, RNNs, Attention & Transformers / 卷积网络、循环网络、注意力与Transformer

Part IV: Generative Models / 生成模型 (下半学期)

Lecture	Topic / 主题
lecture4	Maximum Likelihood & MLE / 最大似然估计
lecture5	Latent Variable Models & ELBO / 潜变量模型与证据下界
LVM_02	VAE & Reparameterization / 变分自编码器与重参数化
NF_01	Normalizing Flow Models (1) / 标准化流模型(1)
lecture8	Normalizing Flow Models (2) / 标准化流模型(2)
lecture9	GANs / 生成对抗网络
lecture10	f-GANs, Wasserstein GAN, BIGAN, CycleGAN / f-散度 GAN, WGAN, 双流GAN, 循环GAN
lecture11	Energy-Based Models / 基于能量的模型
lecture12	Score Matching & NCE / 分数匹配与噪声对比估计
lecture13	Score-Based Models (1) / 基于分数的模型(1)
lecture14	Score-Based Models (2) / 基于分数的模型(2)
lecture15	Diffusion Models (DDPM, SDE) / 扩散模型
lecture16	Diffusion Models: SDE+ODE, Architecture / 扩散模型进阶

2. Core Concepts / 核心概念梳理

2.1 Image Filtering / 图像滤波

Convolution / 卷积

- Definition (定义): Replace each pixel with a weighted sum of its neighbors using a kernel. / 用卷积核将每个像素替换为其邻域的加权。
- Formula (公式): $G[i,j]=\sum H[m,n]*F[i-m,j-n]$
- Key property (关键性质): Convolution is associative and commutative; convolution with self is another Gaussian. / 卷积满足结合律和交换律。

Gaussian Filter / 高斯滤波器

- Definition (定义): A low-pass filter that removes high-frequency components. / 去除高频成分的低调滤波器。
- Property (性质): Convoluting two Gaussians of σ_1 and σ_2 yields a Gaussian of $\sigma=\sigma_1^2+\sigma_2^2$. / 两个高斯卷积仍为高斯。

Sharpening Filter / 锐化滤波器

- Definition (定义): original + (original - blurred) = 2*original - blurred. / 原图加上(原图减模糊图)实现锐化。
- Also called (亦称): High-pass filter / 高通滤波器。

2.2 Edge Detection / 边缘检测

Image Gradient / 图像梯度

- Definition (定义): $\nabla f=(\partial f/\partial x,\partial f/\partial y)$, points in direction of most rapid increase. / 指向强度增长最快的方向。
- Gradient magnitude (梯度幅值): $\|\nabla f\|=(\partial f/\partial x)^2+(\partial f/\partial y)^2$
- Gradient direction (梯度方向): $\theta=\text{atan2}(\partial f/\partial y,\partial f/\partial x)$

Sobel Operator / Sobel算子

- Definition (定义): Common approximation of derivative of Gaussian. / 高斯导数的常用近似。
- Kernels (核):
 - x-direction: $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$
 - y-direction: $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

Canny Edge Detector / Canny边缘检测器

- Filter image with derivative of Gaussian / 用高斯导数滤波
- Find magnitude and orientation of gradient / 求梯度幅值和方向
- Non-maximum suppression / 非极大值抑制
- Hysteresis thresholding (two thresholds: high T, low t / 滞后阈值化
- Strong edges: $R > T$; Weak edges: $t < R < T$ (kept only if connected to strong edge) / 强边缘保留，弱边缘仅在与强边缘相连时保留

Scale Space / 尺度空间

- Properties (性质): Edge position may shift with g ; two edges may merge; an edge cannot split into two. / 边缘位置可能随 σ 偏移，可合并但不能分裂。

2.3 Image Resampling / 图像重采样

Aliasing / 混叠

- Definition (定义): Occurs when sampling rate $< 2 \times$ max frequency (Nyquist rate). / 采样率低于奈奎斯特率时发生。
 - Solution (解决方案): Gaussian pre-filtering before subsampling. / 下采样前先做高斯预滤波。
- Gaussian Pyramid / 高斯金字塔
- Definition (定义): Repeatedly blur and downsample by $2 \times$. / 反复模糊后2倍下采样。
 - Space (空间): Total space = $4/3 \times$ original image. / 总空间为原图的4/3。

Interpolation Methods / 插值方法

- Nearest-neighbor (最近值): Fastest, but blocky. / 最快但产生方块效应。
- Bilinear (双线性): Better quality, tent function. / 质量更好。
- Bicubic (双三次): Best quality among common choices. / 常用方法中质量最好。

2.4 Harris Corner Detection / Harris角点检测

SSD Error / SSD误差

- Formula (公式): $E(u,v)=\sum x_i y_i \sum W(w,x_i)[(x_i+u,y_i+v)-(x_i,y_i)]^2$
- Small motion approximation (小运动近似): $E(u,v)=\sum u^2 H_{xx} + 2uv H_{xy} + v^2 H_{yy}$, where H is the second moment matrix. / 二阶矩矩阵H近似。

Second Moment Matrix / 二阶矩矩阵

- Definition (定义): $H=\sum w(x,y)[x^2 \ 2xy \ xy^2]$
- Eigenvalues (特征值): λ_1, λ_2 determine corner/edge/flat classification. / 决定角点/边缘/平坦区域分类。

Classification / 分类

Case (情况)	λ_1, λ_2	Classification (分类)
Both small (都很小)	$\lambda_1 \approx \lambda_2 \approx 0$	Flat / 平坦区域
One large, one small (一大一小)	$\lambda_1 \gg \lambda_2$ or $\lambda_2 \gg \lambda_1$	Edge / 边缘
Both large (都很大)	$\lambda_1 \approx \lambda_2 \gg 0$	Corner / 角点

Harris Operator / Harris算子

- Formula (公式): $R=\det(H)-\frac{\alpha}{4}\text{trace}(H)^2=\lambda_1\lambda_2-\alpha(\lambda_1+\lambda_2)^2$
- Typical α (典型参数): $\alpha \in [0.04, 0.06]$

Harris Detector Steps / Harris检测步骤

- Compute image gradients / 计算图像梯度
- Square of derivatives, Gaussian filter / 梯度平方后高斯滤波
- Compute cornerness function / 计算角点响应函数
- Non-maxima suppression / 非极大值抑制

2.5 Feature Invariance / 特征不变性

Invariance vs. Equivariance / 不变性与等变性

- Invariance (不变性): Feature locations don't change after transformation. / 变换后特征位置不变。
- Equivariance (等变性): Features detected at corresponding locations in transformed images. / 变换后特征在对应位置被检测到。

Harris Invariance Properties / Harris不变性

Transformation (变换)	Property (性质)
Translation (平移)	Equivariant / 等变
Rotation (旋转)	Equivariant (eigenvalues unchanged) / 等变
Affine intensity (仿射强度)	Partially invariant / 部分不变
Scale (尺度)	NOT invariant / 不变

Scale Invariant Detection / 尺度不变检测

- Key idea (核心思想): Find local maximum in both position AND scale. / 在位置和尺度上同时寻找局部极大值。
 - Laplacian of Gaussian (LoG): Blob detector, find maxima in position-scale space. / 斑点检测器，在位置-尺度空间找极大值。
 - Difference of Gaussians (DoG): Approximation of LoG, used in SIFT. / LoG的近似，SIFT中使用。
- Characteristic Scale / 特征尺度
- Definition (定义): Scale producing peak LoG response. / 产生LoG峰值响应的尺度。

2.6 Feature Descriptors & Matching / 特征描述子与匹配

SIFT Descriptor / SIFT描述子

- Process (过程):
 - Take 16×16 window around feature / 取 16×16 窗口
 - Compute gradient orientation per pixel / 计算每像素梯度方向
 - Threshold weak edges / 阈值化弱边缘
 - Create 4×4 grid of cells, 8 orientation bins each / 4×4 网格，每格8方向bin
- Result: 128-dimensional descriptor / 结果为128维描述子
- Properties (性质): Robust to viewpoint (up to -60°), illumination changes. / 对视点变化 (-60° 以内) 和光照变化鲁棒。

HOG Descriptor / HOG描述子

- Steps: Gradient computation $\rightarrow$ Cell histograms (9 bins) $\rightarrow$ Block normalization (L2-norm). / 梯度计算 $\rightarrow$ 单元直方图 $\rightarrow$ 块归一化。

ORB Descriptor / ORB描述子

- Steps: oFAST keypoint detection $\rightarrow$ rBRIEF binary descriptor (256 bits). / oFAST关键点检测 $\rightarrow$ rBRIEF二进制描述子。

Feature Matching / 特征匹配

- L2 distance: $\|f_1-f_2\|$
- Ratio distance: $\|f_1-f_2\| / \|f_1\| - \|f_2\|$ ($f_1(f_2) = \text{best match}, f_2(f_2) = 2 \times \text{best}$) / 比值距离
- ROC Curve: True positive rate vs. false positive rate; AUC summarizes performance. / ROC曲线与AUC指标。

2.7 Image Transformations / 图像变换

Transformation Hierarchy / 变换层级

Transform (变换)	Matrix (矩阵)	DOF	Properties (性质)
Translation (平移)	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	2	Origin may shift / 原点可移动
Euclidean (欧氏)	$R + t$	3	Preserves distances / 保持距离
Similarity (相似)	$sR + t$	4	Preserves angles / 保持角度
Affine (仿射)	$\begin{bmatrix} a & b & 0 \\ 0 & 1 \end{bmatrix}$	6	Parallel lines preserved / 平行线保持
Projective (射影/单应)	General 3×3	8	Lines preserved, parallel not / 直线保持，平行不一定

Homography / 单应变换

- Definition (定义): 8-DOF projective transformation; defined up to scale. / 8自由度数射影变换，由尺度因子决定。
- Minimum matches (最少匹配): 4 point correspondences. / 4对匹配点。

Forward vs. Inverse Warping / 前向与逆向变形

- Forward warping: Send source pixels to target; may cause holes. / 源像素送到目标，可能产生空洞。
- Inverse warping: For each target pixel, sample from source via $T^{-1}()$; use interpolation. / 对每个目标像素从源采样，需插值。

2.8 Image Alignment & Least Squares / 图像对齐与最小二乘

Least Squares / 最小二乘法

- Translation (平移): $t = \text{mean displacement}$; normal equations: $ATA=t=Atb$. / 平移的最优解为平均位移。
- Affine (仿射): 6 unknowns, need ≥ 3 matches. / 6个未知量，至少3对匹配。
- Homography (单应): Solve via SVD; $h = \text{eigenvector of } ATA$ with smallest eigenvalue. / 通过SVD求解。

2.9 RANSAC / 随机抽样一致

Algorithm / 算法步骤

- Randomly select s samples (minimum to fit model) / 随机选 s 个样本
- Fit model / 拟合模型
- Count inliers (within threshold ϵ) / 计数内点
- Repeat N times / 重复 N 次
- Choose model with most inliers / 选择内点最多的模型
- Refit using all inliers (least squares) / 用所有内点重新拟合

Number of Rounds / 迭代次数

- Formula (公式): $N=\log(1-p)/log(1-(1-e)^s)$
- $p = \text{desired probability of success}$ / 期望成功概率
- $e = \text{outlier ratio}$ / 外点比例
- $s = \text{minimum sample size}$ / 最小样本数

Pros & Cons / 优缺点

- Pros: Simple, general, often works well. / 简单、通用、效果好。
- Cons: Parameters to tune; can fail for very low inlier ratios. / 需调参；内点率极低时可能失败。

2.10 Cameras & Projection / 相机与投影

Pinhole Camera / 针孔相机

- Projection (投影): $(x,y,z) \rightarrow (f \cdot x/z, f \cdot y/z)$ using similar triangles. / 利用相似三角形投影。
- Perspective projection matrix / 透视投影矩阵: $K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$

Camera Parameters / 相机参数

- Intrinsics (内参): focal length f , principal point (c_x, c_y) , aspect ratio a , skew s . / 焦距、主点、长宽比、倾斜。
- Extrinsics (外参): Rotation $R(3 \times 3)$ and translation $t(3 \times 1)$. / 旋转矩阵和平移向量。
- Full projection (完整投影): $px=K[R|t]$ / $t \times \text{World}$

Projection Types / 投影类型

- Perspective (透视): General case, parallel lines converge. / 一般情况，平行线汇聚。
- Orthographic (正交): Distance to image plane = ∞ . $(x,y,z) \rightarrow (x,y)$. / 距离无限远。
- Weak perspective (弱透视): Scaled orthographic; good for small depth range. / 缩放正交投影。

Radial Distortion / 径向畸变

- Types: Barrel (桶形) vs. Pincushion (枕形)
- Model: $r_{corrected}=r(1+k_1r^2+k_2r^4+\dots)$

2.11 Stereo Vision / 立体视觉

Disparity & Depth / 视差与深度

- Disparity (视差): $d=x_1-x_2$ (horizontal pixel difference between views). / 左右视图水平像素差。
- Depth (深度): $Z=f \cdot B/d$, where $B = \text{baseline}$, $f = \text{focal length}$. / 深度与视差成反比。

Stereo Matching / 立体匹配

- SSD matching: Sum of squared differences within a window. / 窗口内平方差之和。
- Window size tradeoff: Small $\rightarrow$ more detail but more noise; Large $\rightarrow$ less noise but less detail. / 窗口大小的权衡。

Stereo as Energy Minimization / 立体视觉作为能量最小化

- Energy: $E(d)=\sum data_cost(x,d(x)) + \lambda \sum smoothness_cost(d(x),d(y))$
- Smoothness costs: Potts model (0 if same, λ if different); L1 distance; / Potts模型: L1距离。

Multi-View Stereo (MVS) / 多视图立体

- SSD: Sum of SSD over multiple baselines. / 多基线SSD之和。
 - Matching scores: SSD, SAD, ZNCC (zero-mean normalized cross-correlation). / 匹配评分方法。
- Plane-Sweep Stereo / 平面扫描立体
- Sweep family of planes at different depths, reproject neighbors, compare. / 在不同深度扫描平面，重投影邻居并比较。
 - NeRF: Represent scenes as $(x,y,z) \rightarrow RGB+q$ functions, volume rendering. / 用神经网络表示场景的辐射场。

2.12 Neural Networks / 神经网络

Linear Classifier / 线性分类器

- Score function (得分函数): $f(x,W)=Wx+b$
- Three viewpoints (三个视角): Algebraic (代数), Geometric (几何), Template matching (模板匹配)

Loss Functions / 损失函数

- Cross-entropy loss (交叉熵损失): $L=-\log(\exp(fy_i)/Z)\exp(fj_i)$
- Softmax (软最大): $P(y=j|x)=\exp(fj)/\sum_k \exp(fk)$

Neural Network Architecture / 神经网络架构

- MLP (多层感知机): Linear functions separated by non-linear activations. / 线性函数间插入非线性激活函数。
- Gradient descent (梯度下降): $W \leftarrow W-\eta \nabla L(W)$; learning rate η crucial. / 学习率 η 至关重要。
- Backpropagation (反向传播): Efficient gradient computation via chain rule. / 通过链式法则高效计算梯度。

2.13 CNNs, RNNs, Attention, Transformers / CNN、RNN、注意力、Transformer

Convolutional Layer / 卷积层

- Output size (输出尺寸): $(N-F)/stride+1$
- Filters (滤波器): Extend full depth of input; produce activation maps. / 滤波器覆盖输入全深度，产生激活图。
- Padding & Stride: Control output spatial dimensions. / 控制输出空间维度。

Pooling Layer / 池化层

- Max pooling (最大池化): Reduces spatial dimensions; operates per activation map. / 减小空间维度，逐激活图操作。

RNN / 循环神经网络

- Recurrence (递推): $h_t=(W \cdot x_t+U \cdot h_{t-1})$; same weights at each time step. / 每个时间步使用相同权重。
- BPTT: Backpropagation Through Time. / 时间反向传播。
- Truncated BPTT: Backpropagate only for limited steps. / 截断时间反向传播。
- Types: Vanilla RNN, LSTM, GRU. / 类型。

Self-Attention / 自注意力

- Query-Key-Value (查询-键-值): Attention $(Q,K,V)=\text{softmax}(QK^T/dk)V$
- Multi-Head Attention (多头注意力): Multiple attention heads in parallel; concatenate outputs. / 多个注意力头并行，拼接输出。

Transformer

- Encoder (编码器): Self-attention + FFN, with residual connections and layer norm. / 自注意力+前馈网络，带残差连接和层归一化。
- Decoder (解码器): Masked multi-head attention (cannot see future) + cross-attention + FFN. / 掩码多头注意力+交叉注意力+前馈网络。
- Positional Encoding (位置编码): Injects sequence order information. / 注入序列顺序信息。

2.14 Maximum Likelihood Estimation / 最大似然估计

KL Divergence / KL散度

- Definition (定义): $DKL(p||q)=\sum p(q)\log(p(q)/q(q)) \geq 0$; asymmetric. / 非对称，衡量两个分布的差异。
- Interpretation (解释): Extra bits needed when using code optimized for q on data from p . / 用 q 的编码压缩来自 p 的数据时多用的比特数。

MLE / 最大似然估计

- Formula (公式): $\theta^*=\text{argmax}_{\theta} \sum x \in \mathcal{D} \log P_\theta(x)$
- Equivalence (等价性): Minimizing $KL(p||p_\theta) \equiv$ maximizing expected log-likelihood. / 最小化KL散度等价于最大化期望对数似然。

Monte Carlo Estimation / 蒙特卡罗估计

- Unbiased (无偏): $E[g]=E[g(x)]$
- Variance (方差): $\text{Var}(g)=\text{Var}(g(x))/T$; reduces with more samples. / 更多样本降低方差。

Bias-Variance Tradeoff / 偏差-方差权衡

- High bias (高偏差): Underfitting (model too simple). / 欠拟合。
- High variance (高方差): Overfitting (model too complex). / 过拟合。
- Regularization (正则化): objective-loss+R(M). / 目标函数加入正则项。

2.15 Latent Variable Models & VAE / 潜变量模型与变分自编码器

Latent Variable Models / 潜变量模型

- Generative process (生成过程): $z \sim p(z)$, $x \sim p(x|z)$. / 先采样潜变量，再由其生成数据。
- Marginal likelihood (边际似然): $p(x;\theta)=\int p(x,z)dz$ — often intractable. / 通常难以计算。

Mixture of Gaussians / 高斯混合模型

- z -Categorical ($1,...,K$): $p(x|z,k)=\mathcal{N}(x|\mu_k,\Sigma_k)$. / 离散潜变量的混合模型。

VAE / 变分自编码器

- Infinite mixture (无限混合): $z \sim \mathcal{Q}(0,1)$; $p(x|z)=\mathcal{N}(x|\mu(z),\Sigma(z))$; $\mu_\theta, \Sigma_\theta$ are neural networks. / 连续潜变量的无限混合高斯。

ELBO / 证据下界

- Formula (公式): $\log p(x) \geq \Sigma q(z;\phi) \log p(x,z;\theta)+H(q)=L(x;\theta,\phi)$
- Equality condition (等号条件): When $q(z)=p(z)$. / $\text{KL}(q||p)=0$. / 当变分分布等于真实后验时等号成立。

Decomposition (分解): $\log p(x)=\text{ELBO}+\text{DKL}(q(z;\phi)||p(z;\theta))$

Autoencoder Perspective / 自编码器视角

- ELBO = Reconstruction term - KL term: $L=E_{q(z|x)}[\log p(x|z;\theta)]-\text{DKL}(q(z|x)||p(z))$
- Reconstruction (重建项): Encourages $x^h \approx x$. / 鼓励重建接近还原。
- KL term (KL项): Forces latent distribution close to prior $p(z)$. / 迫使潜变量分布接近先验。

Reparameterization Trick / 重参数化技巧

- Key idea (核心思想): Sample $\epsilon \sim \mathcal{N}(0,1)$, then $z=\mu+\sigma\epsilon$. / 将随机性转移到 ϵ ，使梯度可通过确定函数后传播。
- Benefit (优势): Enables gradient-based optimization; lower variance than REINFORCE. / 实现梯度优化，方差低于REINFORCE。

Amortized Inference / 摊销推断

- Idea (思想): Learn $q_\phi(z|x)=\text{encoder}(x)$ to map each x to variational parameters. / 学习编码器网络将 x 映射到变分参数，避免逐点优化。

2.16 Normalizing Flow Models / 标准化流模型

Core Idea / 核心思想

- Make $q_\theta(z)$ deterministic and invertible; then $p(x)$ is tractable via change of variables. / 使映射确定且可逆，通过变量替换求 $p(x)$ 计算量。

Change of Variables / 变量替换公式

- 1D: $p(x)=p(z(x))|h'(x)|$, where $h=f^{-1}$. / 一维情况。
- General: $p(x)=p(z(f^{-1}(x)))|\det(\partial f^{-1}/\partial x)|=p(z)|\det(\partial f/\partial z)|^{-1}$. / 一般情况。

Flow of Transformations / 变换流

- Composition (组合): $z_0 \rightarrow f_1 \rightarrow z_1 \rightarrow f_2 \rightarrow \dots \rightarrow f_M \rightarrow z_M=x$

- Likelihood: $p(x)=p(z(0))\prod_{t=1}^M |\det(\partial f_t/\partial z_t)|$

2.20 BiGAN / 双向生成对抗网络

Bidirectional GAN (BiGAN) / 双向GAN

- Motivation (动机): Standard GANs learn a generator $G_{z \rightarrow x}$ but cannot infer the latent representation z given x . / 标准GAN只学到生成器, 无法推断潜在表示。
- Architecture (架构): Add an encoder $E: x \rightarrow z$ alongside the generator $G: z \rightarrow x$; discriminator D takes (x, z) pairs and distinguishes $(x, E(x))$ from $(G(z), z)$. / 增加编码器 E . 判别器在 (x, z) 对上区编码对与生成对。
- Objective (目标):
 - $\min_G \max_D V(D, E, G) = E_{x \sim p_{data}} [\log D(x, E(x))] + E_{z \sim p(z)} [\log(1 - D(G(z), z))]$
- Result (结果): At equilibrium, $E = G^{-1}$ (encoder inverts the generator). / 均衡时编码器为生成器的逆。
- ALI (Adversarially Learned Inference): Simultaneous proposal, equivalent to BiGAN. / 同时提出的等价方法。

2.21 CycleGAN / 循环生成对抗网络

Image-to-Image Translation / 图像到图像翻译

- Problem (问题): Given images from two domains X and Y , learn to translate between them. / 给定两个域的图像, 学习在域间翻译。
- Paired vs. Unpaired (配对与非配对): Paired examples are expensive; CycleGAN works with unpaired data. / 配对数据昂贵, CycleGAN使用非配对数据。
- CycleGAN Architecture / CycleGAN架构
 - Generators (生成器): $G_X: X \rightarrow Y$ and $F_Y: X \rightarrow X$, 双向生成器。
 - Discriminators (判别器): D_Y distinguishes real Y from $G(X)$; D_X distinguishes real X from $F(Y)$. / 双判别器。
 - Adversarial loss (对抗损失): $\mathcal{L}_{GAN}(G, D_Y, X, Y)$
 - $\pm \mathcal{L}_{GAN}(F, D_X, X, Y)$ / 双向对抗损失。
 - Cycle consistency loss (循环一致性损失): $\lambda \|E(X) - F(G(X)) - X\|_1 + \|E(Y) - F(G(Y)) - Y\|_1$. / 确保往返循环一致。
 - Overall objective (总目标): $\min_F, G, D_X, D_Y \mathcal{L}_{GAN} + \lambda \cdot \mathcal{L}_{cyc}$.

AlignFlow / 对齐流

- Key idea (核心思想): If G is a normalizing flow model, $F = G^{-1}$ is free; exactly cycle-consistent. / 若生成器为流模型, 逆向自然可得, 精确循环一致。
- Training (训练): Can train via MLE and/or adversarial learning. / 可通过MLE和/或对抗机器学习。

StarGAN / 星形GAN

- Extension (扩展): Handles multiple domains with a single generator. / 单一生成器处理多个域的解释。

2.22 Energy-Based Models / 基于能量的模型

EBM Definition / EBM公式

- Definition (定义): $p_\theta(x) = \exp(\theta(x))Z(\theta) = 1/Z(\theta) \exp(\theta(x))$, where $\theta(x)$ is any function (e.g., neural network) and $Z(\theta) = \int \exp(\theta(x))dx$ is the partition function. / 任意函数取指数后归一化, 分母为配分函数。
- Key property (关键性质): Very flexible architecture — θ can be any neural network (no invertibility, factorization, or latent variable constraints). / 架构极其灵活, 可为任意神经网络。
- Partition function (配分函数): $Z(\theta) = \int \exp(\theta(x))dx$ — generally intractable. / 通常不可计算。

Energy Function / 能量函数

- Interpretation (解释): $-\theta(x)$ is the “energy”; low energy = high probability. / 负值为“能量”, 低能量对应高概率。
- Non-negativity trick (非负技巧): Given any function θ , ensure non-negativity via $g(\theta(x)) = \theta(x)^2$. / $\theta(x)$ 的平方, 或 $\log(1 + \exp(\theta(x)))$. / 通过平方、指数、绝对值等保证非负。

Maximum Likelihood Training / 最大似然训练

- Log-likelihood (对数似然): $\log p_\theta(x) = \theta(x) - \log Z(\theta)$. / 对数似然等于函数值减去配分函数对数。
- Gradient (梯度): $\nabla_{\theta} \log p_\theta(x) = \nabla \theta f(x)$ (train) - $E_{p_\theta}[\nabla \theta f(x)]$ / 梯度=正相位 (训练: 训练数据期望能量) 减负相位 (拉低模型采样能量)。
- Positive phase vs. Negative phase (正相位与负相位):
 - Positive phase (正相位): $\nabla \theta f(x)$ increases probability of training data. / 提高训练数据概率。
 - Negative phase (负相位): $E_{p_\theta}[\nabla \theta f(x)]$ decreases probability of model samples. / 降低模型样本概率。

Contrastive Divergence / 对比度散度

- Algorithm (算法): Instead of computing $Z(\theta)$ exactly, use Monte Carlo estimate. / 用蒙特卡洛估计代替精确计算配分函数。
- Update step (更新步骤): $\nabla \theta f_\theta(x) - \nabla \theta f_\theta(x_{\text{sample}})$, where $x_{\text{sample}} \sim p_\theta$. / 用模型采样的梯度近似负相位。
- Key idea (核心思想): Make training data more likely than typical samples from the model. / 使训练数据比模型典型样本概率更高。

Sampling from Models / 从EBM采样

- MCMC (马尔可夫链蒙特卡洛):
 - Initialize x_0 randomly / 随机初始化
 - Propose $x' = x_t + \text{noise}$ / 提议新样本
 - If $\theta(x') > \theta(x_t)$: accept; else accept with probability $\exp(\theta(x') - \theta(x_t))$ / 接受或以一定概率接受
 - Converges to p_θ , but can be slow / 收敛但可能很慢
- Langevin MCMC (朗之万MCMC): $xt+1=xt+e+\nabla\theta\log p_\theta(xt)+2e_{zt}$, where $zt\sim\mathcal{N}(0,1)$. / 利用梯度信息加速采样的MCMC方法。
- Score function (分数函数): For EBMs, $\nabla\log p_\theta(x)=\nabla\theta f(x)$ (since $\nabla\log Z(\theta)=0$). / EBM的分数函数 (对数密度对x的梯度) 恰好等于 θ 的梯度。

Score-Based Generative Modeling / 基于分数的生成模型

- Key idea (核心思想): Directly estimate the score function $\nabla\log p(x)$ (using a neural network $s_\theta(x) \approx \nabla\log p(x)$. / 直接用神经网络估计分数函数。
 - Score matching (分数匹配): Train s_θ by minimizing $E[pdata]{\|s_\theta(x)-\nabla\log pdata(x)\|^2}$ (or variants avoiding $\nabla\log pdata(x)$). / 通过分数匹配目标训练。
 - Denoising Score Matching (去噪分数匹配): Add noise α to data, then match the score of the perturbed distribution. / 对数据加噪后匹配扰动分布的分数。
 - Noise Conditional Score Network (NCSN): Use multiple noise scales $\{\alpha_1>\alpha_2>\dots>\alpha_L\}$; train one network conditioned on noise level. / 使用多尺度噪声, 训练条件分数网络。
 - Annealed Langevin Dynamics (退火朗之万动力学): Sequentially sample with decreasing noise levels; each level initialized from previous. / 逐步降低噪声水平采样。
- #### EBM vs. Other Generative Models / EBM与其他生成模型比较
- vs. Autoregressive/Flow/VAE: EBM has much more flexible architecture (no restrictions on θ), but sampling is slower (requires MCMC). / EBM架构更灵活, 但采样需要MCMC。
 - vs. GAN: EBM provides stable training and tractable (non-annealed) likelihood; GAN has fast sampling but unstable training and intractable likelihood. / EBM训练稳定, GAN采样快但训练不稳定。
 - Composability (可组合性): $p_{\text{combined}}(x) \propto \Pi p_i(x)$ — can combine multiple EBMs by multiplying their densities. / 多个EBM可乘性组合, 适用于零样本组合推理。

3. Key Formulas & Theorems / 重点公式与定理

3.1 Convolution / 卷积

- Formula: $G[i,j]=\sum_m\sum_n\{I[m,n]\cdot F[i-m,j-n]$
- Variables: H = kernel, F = input image, G = output image
- Conditions: Kernel size determines neighborhood / 核大小决定邻域范围

3.2 Gaussian Filter / 高斯滤波

- Formula: $G(x,y)=1/2\pi\sigma^2\exp(-x^2/2-y^2/2\sigma^2)$
- Variables: σ isigma = standard deviation (scale) / 标准差 (尺度)
- Conditions: Larger σ = more blurring / σ 越大模糊程度

3.3 Image Gradient / 图像梯度

- Formula: $\nabla f=[\partial f/\partial x,\partial f/\partial y];\|\nabla f\|=(\partial f/\partial x)^2+(\partial f/\partial y)^2$
- Variables: ∂ partial $\partial f/\partial x$, ∂ partial $\partial f/\partial y$ = partial derivatives / 偏导数
- Conditions: Computed via finite differences or derivative of Gaussian / 通过有限差分或高斯导数计算

3.4 Harris Corner Response / Harris角点响应

- Formula: $R=\det(H)-\frac{1}{4}\text{trace}(H)^2=1/2\lambda_1\lambda_2-a(1+\lambda_1)^2$
 - Variables: H = second moment matrix, $a\in[0.04,0.06]$. λ_1,λ_2 = eigenvalues
 - Conditions: $R>0$ — corner; $R\approx 0$ — edge; $R<0$ — flat (using alternate sign convention)
- ### 3.5 Perspective Projection / 透视投影
- Formula: $(x,y,z)\rightarrow (fx/z+cx,fy/z+cy)$
 - Variables: f = focal length, (cx,cy) = principal point
 - Conditions: Pinhole camera model / 针孔相机模型
- ### 3.6 Stereo Depth / 立体深度
- Formula: $Z=f/B/d$
 - Variables: Z = depth, f = focal length, B = baseline, d = disparity
 - Conditions: Rectified stereo pair / 校正后的立体对

3.7 RANSAC Iterations / RANSAC迭代次数

- Formula: $N=\log(1-p)/\log(1-(1-e)^s)$
 - Variables: p = success probability, e = outlier ratio, s = min sample size
 - Conditions: Assumes < 50% outliers / 假设外点比例 < 50%
- ### 3.8 ELBO / 证据下界
- Formula: $\log p(x;\theta)\geq E_q[\log p(x,z;\theta)]-E_q[\log q(z;\varphi)]=E_q[\log p(x,z;\theta)-DKL(q\varphi(z)||p(z))]$
 - Variables: q = variational distribution, θ = model params, φ = variational params
 - Conditions: Equality when $q(z|\varphi)=p(z|x;\theta)$ / 当变分分布等于后验时等号成立
- ### 3.9 Change of Variables (Flow) / 变量替换 (标准化流)
- Formula: $p(x)=p(z=f(x))\cdot|\det(\partial f^{-1}/\partial x)|$
 - Variables: f = invertible transformation, $x,z\in\mathbb{R}^n$ (same dimension)
 - Conditions: f must be invertible; x,z continuous / f 必须可逆; x,z 连续且同维

3.10 GAN Objective / GAN目标

- Formula: $\min_G\max_D\mathcal{L}_{GAN}(\log p(x)) + E_{z\sim p(z)}[\log(1-D(G(z)))]$
- Variables: G = generator, D = discriminator, p_z = prior on z
- Conditions: At optimum, equivalent to minimizing JSD / 最优时等价于最小化JSD数

3.11 Cross-Entropy Loss / 交叉熵损失

- Formula: $L=-\log(\exp(f(y))/\sum_j\exp(f_j))$
- Variables: f_j = score for class j , y_i = true label
- Conditions: Most common classification loss / 最常用的分类损失

3.12 Conv Output Size / 卷积输出尺寸

- Formula: $\text{output}=(N-F+2)\text{Stride}+1$
 - Variables: N = input size, F = filter size, P = padding
 - Conditions: Must be integer / 必须为整数
- ### 3.13 f-Divergence / f-散度
- Formula: $D_f(p,q)=E_x[-q(f(p(x),q(x)))]$, where f is convex with $f(1)=0$
 - Variables: p,q = probability distributions, f = convex generator function
 - Conditions: $D_f(p,q)\geq 0$; equals 0 iff $p=q$ / 非负, 等号当且仅当两分布相同

3.14 Wasserstein Distance / Wasserstein距离

- Formula: $W(p,r,p_g)=\sup\{f\mid f\leq 1\text{ and }E_x[-pr(f(x))] = E_x[-pg(f(x))]$
- Variables: p,r = real and generated distributions, f = 1-Lipschitz function
- Conditions: Meaningful even for non-overlapping distributions / 即使分布不重叠也有意义

3.15 EBM Probability / EBM概率

- Formula: $p_\theta(x)=1/Z(\theta)\exp(\theta(x))$, where $Z(\theta)=\int\exp(\theta f(x))dx$
 - Variables: $\theta f(x)$ = any function (energy negative), $Z(\theta)$ = partition function
 - Conditions: $Z(\theta)$ generally intractable / 配分函数通常不可计算
- ### 3.16 EBM Log-Likelihood Gradient / EBM对数似然梯度
- Formula: $\nabla\log p_\theta(x)=\nabla\theta f(x)$ (train)- $E_{p_\theta}[\nabla\theta f(x)]$
 - Variables: First term = positive phase, second term = negative phase
 - Conditions: Negative phase estimated via MCMC samples / 负相位通过MCMC样本估计

3.17 Langevin MCMC / 朗之万MCMC

- Formula: $xt+1=xt+e+\nabla\theta f_\theta(xt)+2e_{zt}$, where $zt\sim\mathcal{N}(0,1)$
 - Variables: e = step size, $\nabla\theta f_\theta(x)$ = score function
 - Conditions: Converges to p_θ as $e\rightarrow 0$, $T\rightarrow\infty$ / 当步长趋于0, 迭代次数趋于无穷时收敛
- ### 4. Typical Questions & Solutions / 典型例题与解答
- #### Q1: Gaussian Filter Properties / 高斯滤波属性
- Problem (问题): If you convolve an image with a Gaussian of σ_1 and then with a Gaussian of σ_2 , what is the effective σ ?
 - Solution (解答): $\sigma_{\text{eff}}=\sqrt{\sigma_1^2+\sigma_2^2}$. This follows from the property that convolution of two Gaussians is another Gaussian, and their variances add. / 两个高斯卷积仍为高斯, 方差相加。
- #### Q2: Nyquist Rate / 奈奎斯特率
- Problem (问题): An image has maximum frequency 50 cycles/pixel. What is the minimum sampling rate to avoid aliasing?
 - Solution (解答): Nyquist rate = $2\times$ max frequency = 100 samples/pixel interval. Sampling rate must be ≥ 100 . / 采样率必须至少为最高频率的两倍。
- #### Q3: Harris Corner Classification / Harris角点分类
- Problem (问题): For a window where the second moment matrix H has eigenvalues $\lambda_1=5000$ and $\lambda_2=50$, what type of region is this?
 - Solution (解答): $\lambda_1\gg\lambda_2\rightarrow$ This is an edge region. The intensity changes significantly in one direction but not the other. / 这是一个边缘区域, 强度在一个方向变化大但在另一方向变化小。
- #### Q4: Homography DOF / 单应变换自由度
- Problem (问题): How many point correspondences are needed to solve for a homography?
 - Solution (解答): A homography has 8 DOF (3x3 matrix up to scale). Each point correspondence gives 2 equations. Therefore, minimum 4 point correspondences are needed. / 单应矩阵8个自由度, 每对匹配提供2个方程, 至少需4对。
- #### Q5: RANSAC Rounds / RANSAC轮数计算
- Problem (问题): For computing a homography (s=4) with 30% outlier ratio and p=0.99, how many RANSAC rounds?
 - Solution (解答): $N=\log(1-0.99)/\log(1-(1-0.3)^4)=\log(0.01)/\log(1-0.74)\approx 17$ rounds. / 约需17轮迭代。
- #### Q6: Stereo Depth Calculation / 立体深度计算
- Problem (问题): Given a stereo system with baseline B=0.2m, focal length f=500 pixels, and disparity $d=50$ pixels, what is the depth Z?
 - Solution (解答): $Z=f\cdot B/d=500\times0.2/50=2.0$ meters. / 深度为2米。
- #### Q7: Conv Layer Output Size / 卷积层输出尺寸
- Problem (问题): Input 32x32x3, 10 filters of size 5x5x3, stride 1, padding 0. What is the output size?
 - Solution (解答): $\text{Output}=(32-5+1)/1+1=28$. So output size is 28x28x10. / 输出尺寸为28x28x10。
- #### Q8: VAE ELBO Decomposition / VAE的ELBO分解
- Problem (问题): Write the ELBO in terms of reconstruction loss and KL divergence.
 - Solution (解答): $\text{ELBO}=\mathbb{E}_q(z|x)[\log p(x|z;\theta)]-DKL(q\varphi(z)||p(z))$
 - First term: reconstruction loss (autoencoding) / 重建损失
 - Second term: KL divergence of approximate posterior from prior / 近后验与先验的KL散度
- #### Q9: Flow Model Likelihood / 标准化流似然计算
- Problem (问题): In a normalizing flow with 3 invertible layers, write the log-likelihood.
 - Solution (解答):

- $\log p(x)=\log\log Z(f^{-1}(x))+2m=13\log\det(\partial f/\partial z)$ (where $z=f^{-1}(x)$ and $zm=fm-(zm+1)$. / 对数似然等于先验对数概率加上各层雅可比行列式之和。

Q10: GAN Optimal Discriminator / GAN最优判别器

- Problem (问题): Derive the optimal discriminator for fixed generator.
- Solution (解答): $D^*(x)=pdata(x)/pdata(x)+pG(x)$
 - When $pG=pdata$: $D^*=1/2$ (cannot distinguish) / 无法区分真假
 - When $pG\neq pdata$: D^* assigns higher probability to whichever distribution has higher density at x . / 判别器给密度更高的分布更高的概率。

Q11: Reparameterization Trick / 重参数化技巧

- Problem (问题): Why can't we directly backpropagate through sampling $z\sim q(z|\varphi)$?
- Solution (解答): Sampling is a stochastic operation — gradients cannot flow through random nodes. Reparameterization: $z=\mu+\alpha\epsilon$ where $\epsilon\sim\mathcal{N}(0,1)$. Now z is a deterministic function of φ and α ; ∇ can flow through μ and α . / 采样是随机操作, 梯度无法通过。重参数化将随机性转移到 α , 使梯度可通过确定函数 μ 和 α 传递。

Q12: Change of Variables (1D) / 变量替换(一维)

- Problem (问题): If $x=U[0,2]$ and $X=4Z$, find $p(X)$.
- Solution (解答): $h(X)=X/4$, $h'(X)=1/4$
 $p(X)=p(X/4)\cdot|1/4|=1/2\times(1/4)=1/8$ for $x\in[0,8]$
- This is uniform on $[0,8]$, as expected. / 结果为 $[0,8]$ 上的均匀分布。

Q13: NICE vs Real-NVP / NICE与Real-NVP区别

- Problem (问题): What is the key difference between NICE and Real-NVP coupling layers?
- Solution (解答):
 - NICE: $xd+1:=zd+1+n+\theta(z1;d)$; $\det(J)=1$ (volume-preserving). / 加性耦合, 体积保持。
 - Real-NVP: $xd+1:=zd+1+n\odot\exp(a\theta(z1;d))+\mu(z1;d)$; $\det(J)=\exp(\sum\alpha)$ (non-volume-preserving). / 仿射耦合, 非体积保持, 表达能力更强。

Q14: MAF vs IAF / 掩码回归流与逆自回归流

- Problem (问题): When would you choose MAF over IAF?
- Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse ($x\rightarrow z$) is parallel. Choose IAF when you need efficient sampling (generation), since its forward ($z\rightarrow x$) is parallel. / MAF适合训练 (需密度估计), IAF适合生成 (需快速采样)。

Q15: Camera Projection Matrix / 相机投影矩阵

- Problem (问题): Write the full projection from world coordinates to pixel coordinates.
- Solution (解答):
$$xpixel=K[R|t]Xworld$$
 where $K=[f_x,c_x,f_y,c_y],[0,0,1]$ (intrinsics), $R=3\times 3$ rotation, $t=3\times 1$ translation (extrinsics). / 内参矩阵 K , 旋转 R 和平移 t 组成完整投影。
- End of CS5187 Review Material / CS5187复习资料完

Q16: WGAN vs. Original GAN / WGAN与原始GAN对比

- Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?
- Solution (解答):
 - Jensen-Shannon divergence: When $pdata$ and pG have non-overlapping supports (common in high dimensions), JSD $\rightarrow\log 2$ is constant — a gradient provides no useful learning signal. / 当分布不重叠时JSD度数为常数, 梯度为零, 无法提供学习信号。
 - Wasserstein distance: Provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / 即使分布不重叠也提供有意义的连续梯度, 损失与样本质量相关。 - Practical benefit: WGAN loss decreases as sample quality improves; can be used as a training monitor. / WGAN损失随样本质量提升而下降, 可作为训练监控指标。

Q17: CycleGAN Cycle Consistency / CycleGAN循环一致性

- Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
- Solution (解答):
 - Without cycle consistency: The adversarial losses alone only ensure that $G(X)$ looks like domain Y and $F(Y)$ looks like domain X , but there is no guarantee that $F(G(X))\approx X$. The generators could map all inputs to a single mode in the target domain (mode collapse). / 仅靠对抗损失无法保证往返一致, 生成器可能将所有输入映射到目标域的单一模式。
 - With cycle consistency: $F(G(X))\approx X$ and $G(F(Y))\approx Y$ enforce that the translation preserves information about the — the round trip must recover the original. / 循环一致性迫使翻译保留信息内容, 往返必须恢复原始输入。 - Loss: $\mathcal{L}_{cyc}=E\|\nabla F(G(X))-X\|_1+E\|Y-\nabla G(F(Y))-Y\|_1$. / 循环一致性损失用L1距离衡量往返差异。

Q18: EBM Gradient Interpretation / EBM梯度解释

- Problem (问题): Explain the positive phase and negative phase in EBM maximum likelihood training.
- Solution (解答):
 - Positive phase (正相位): Pushes up θ at the training data location, making training data more probable. / 在训练数据位置抬高 θ , 提高训练数据概率。
 - Negative phase (负相位): Pushes down θ at locations where the model currently assigns high probability (model samples), reducing probability of non-data regions. / 在模型当前高概率位置拉低 θ , 降低非数据区域的概率。
 - Contrastive divergence: Approximates the negative phase with MCMC samples: $\nabla\theta f_\theta(x)-\nabla\theta f_\theta(x_{\text{sample}})$. / 用MCMC采样近似负相位。

Q19: Score Function in EBMs / EBM中的分数函数

- Problem (问题): Why is the score function $\nabla\log p_\theta(x)$ tractable for EBMs?
- Solution (解答):
 - $\nabla\log p_\theta(x)=\nabla[x\theta f(x)-\log Z(\theta)]=\nabla\theta f(x)-0=\nabla\theta f(x)$ since $Z(\theta)=\int\exp(\theta f(x))dx$ does not depend on x . its gradient w.r.t. x is zero. Therefore, the score function depends only on $\theta f(x)$, which is easy to compute (just backprop through the network). / 配分函数不依赖于 x , 其对的梯度为零, 因此分数函数只依赖于 $\theta f(x)$, 可通过神经网络反向传播轻松计算。

Q20: Score Matching vs MLE / 分数匹配 vs 最大似然

- Problem (问题): Why is score matching preferred over MLE for training EBMs?
- Solution (解答):
 - MLE requires computing $\log Z(\theta)$ and its gradient, which involves intractable integrals over high-dimensional spaces. Score matching sidesteps this by directly matching the score function $\nabla\log p_\theta(x)=\nabla\theta f(x)$, which is independent of $Z(\theta)$. This avoids MCMC sampling during training. / MLE需计算不可解的配分函数, 而分数匹配直接匹配配分函数导数 (不依赖配分函数), 避免了训练中的MCMC采样。

Q21: DDPM Sampling Process / DDPM采样过程

- Problem (问题): Describe the generation process of a trained DDPM model.
- Solution (解答):
 - Sample $xT\sim\mathcal{N}(0,1)$ (pure noise) / 从标准高斯采样纯噪声
 - For $t=T-1, \dots, 1$: / 从 $T-1$ 迭代: - Predict noise: $\epsilon_t=e_\theta(x,t,t)$ / 预测噪声 - Compute mean: $\mu_t=1-\beta_t(x-\beta_{t-1}\epsilon_t)$ / 计算均值 - Sample: $x_t=x_{t-1}+\sigma_t\epsilon_t$, where $z\sim\mathcal{N}(0,1)$ (except at $t=1$) / 采样 3. Return x_0 / 返回最终结果

Q22: SDE vs ODE for Generation / SDE vs ODE生成

- Problem (问题): Compare using the reverse SDE vs. the probability flow ODE for sampling.
- Solution (解答):
 - Reverse SDE: Stochastic; adds noise at each step; produces diverse samples; slower convergence. / 随机性; 逐步加噪; 产生多样化样本; 收敛较慢。
 - Probability Flow ODE: Deterministic; unique mapping from noise to data; enables exact likelihood computation; can use adaptive step-size ODE solvers for speed. / 确定性; 噪声到数据的唯一映射; 可确定计算似然; 可使用自适应步长ODE solvers for speed. / 确定性; 噪声到数据的唯一映射; 可确定计算似然; 可使用自适应步长ODE求解器加速。

- size ODE solvers for speed. / 确定性; 噪声到数据的唯一映射; 可确定计算似然; 可使用自适应步长ODE求解器加速。
- Both have the same marginal distributions $p_t(x)$ at all times. / 两者在所有时间有相同的边际分布。

Q23: NCE vs Score Matching / NCE vs 分数匹配

- Problem (问题): Compare NCE and Score Matching as training methods for EBMs.
- Solution (解答):

	NCE	Score Matching
Approach / 方法	Binary classification (data vs noise) / 二分类	Minimize Fisher divergence / 最小化Fisher散度
Requires / 需要	Choice of noise distribution p_n / 选择噪声分布	Score function computation / 分数函数计算
Partition function / 配分函数	Learned as a parameter / 作为参数学习	Not needed / 不需要
Limitation / 限制	Noise must cover data support / 噪声须覆盖数据支持	Hessian trace (solved by DSM) / Hessian迹 (DSM解决)

Part V: Score Matching & Diffusion Models / 分数匹配与扩散模型 (lecture12-16)

2.21 Score Matching / 分数匹配

- Score Function / 分数函数
 - Definition (定义): $s_\theta(x) = \nabla\log p_\theta(x)$. For EBMs, $s_\theta(x) = \nabla\theta f(x)$ since the partition function $Z(\theta)$ does not depend on x . / 对数概率密度对 x 的梯度。 EBM中分数函数与配分函数无关。
 - Key Points (要点): Does not require knowledge of $Z(\theta)$; enables training without computing normalizing constants. / 无需计算归一化常数即可训练。
- Fisher Divergence / Fisher散度
 - Formula (公式): $D_F(p||q)=\frac{1}{2}E_{p||q}[||p(x)-s_\theta(x)||^2]$
 - Key Points (要点): Measures distance between two distributions via their score functions. / 通过分数函数数量两个分布的距离。
- Score Matching Objective / 分数匹配目标
 - Formula (公式): $\mathcal{L}(\theta)=E_{p||q}[\|s_\theta(x)-\nabla\log p_q(x)\|^2]$
 - Key Points (要点): Equivalent to Fisher divergence up to a constant; requires computing Hessian trace (expensive for deep networks). / 等价于Fisher散度加常数; 需计算Hessian迹。

- Denoising Score Matching (DSM) / 去噪分数匹配
 - Definition (定义): Match score of noise-perturbed distribution: $q(x)=\int pdata(x)\mathcal{N}(x-x,\sigma^2)dx$. / 匹配噪声扰动分布的分数。
 - Formula (公式): $JDSM=E_{x,\epsilon}[\|s_\theta(x-\epsilon)-(x-x)/\sigma^2\|^2]$
 - Key Points (要点): No Hessian needed; is simpler ($x-x)/\sigma^2$; equivalent to predicting noise. / 无需Hessian; 等价于预测噪声。

- 2.22 Noise Contrastive Estimation (NCE) / 噪声对比估计
 - Definition (定义): Train discriminator to distinguish data from noise, implicitly learning the unnormalized density. / 训练判别器区分数据与噪声, 隐式学习非归一化密度。
 - Discriminator (判别器): $D_\theta Z(x)=\theta f_\theta(x)/Z-\nabla p_n(x)$
 - NCE vs GAN: Both use discriminator + cross-entropy. NCE explicitly learns density ratio; GAN uses a separate generator network. / NCE显式学习密度比; GAN使用独立生成网络。

- 2.23 Score-Based Generative Models (NCSN) / 基于分数的生成模型
 - Score-Based Model / 分数模型
 - Definition (定义): Neural network $s_\theta(x)$ trained to approximate $\nabla\log pdata(x)$, without modeling $p(x)$ itself. / 直接近似数据分数函数的神经网络。
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